# ClipSAT : Cambridge & Edexcel Additional Mathematics 9606/4PM1 [ANSWER KEY]

## Unit 7.01: Differentiation & Rates of Change

**Q1.** Find the equation of the tangent line to the curve y = 2x + 1^3 at the point where x = 0 .

[ ] A: y = 2x + 1
[ ] B: y = 6x - 1
**[x] C: y = 6x + 1**
[ ] D: y = 6x + 3

Solution: **Mark Scheme CAIE / Edexcel Grading Steps:** - **Step 1 Method & Formulation:** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]** - **Step 2 Derivation & Calculation:** At x = 0 , y = 1^3 = 1 . Derivative: frac{dy}{dx} = 32x+1^2 cdot 2 = 62x+1^2 . At x = 0 , slope m = 6 . Equation: y - 1 = 6x - 0 implies y = 6x + 1 . **[M1]** - **Step 3 Final Accuracy:** Verified exact result: y = 6x + 1 . **[A1]**

**Q2.** Find the equation of the tangent line to the curve y = 3x + 1^3 at the point where x = 0 .

[ ] A: y = 9x + 3
[ ] B: y = 9x - 1
[ ] C: y = 3x + 1
**[x] D: y = 9x + 1**

Solution: At x = 0 , y = 1^3 = 1 . Derivative: frac{dy}{dx} = 33x+1^2 cdot 3 = 93x+1^2 . At x = 0 , slope m = 9 . Equation: y - 1 = 9x - 0 implies y = 9x + 1 . [M1][A1]

**Q3.** Find the equation of the tangent line to the curve y = 4x + 1^3 at the point where x = 0 .

[ ] A: y = 12x + 3
[ ] B: y = 12x - 1
[ ] C: y = 4x + 1
**[x] D: y = 12x + 1**

Solution: At x = 0 , y = 1^3 = 1 . Derivative: frac{dy}{dx} = 34x+1^2 cdot 4 = 124x+1^2 . At x = 0 , slope m = 12 . Equation: y - 1 = 12x - 0 implies y = 12x + 1 . [M1][A1]

**Q4.** Find the equation of the tangent line to the curve y = 5x + 1^3 at the point where x = 0 .

[ ] A: y = 15x + 3
[ ] B: y = 15x - 1
[ ] C: y = 5x + 1
**[x] D: y = 15x + 1**

Solution: At x = 0 , y = 1^3 = 1 . Derivative: frac{dy}{dx} = 35x+1^2 cdot 5 = 155x+1^2 . At x = 0 , slope m = 15 . Equation: y - 1 = 15x - 0 implies y = 15x + 1 . [M1][A1]

**Q5.** Find the equation of the tangent line to the curve y = 6x + 1^3 at the point where x = 0 .

**[x] A: y = 18x + 1**
[ ] B: y = 6x + 1
[ ] C: y = 18x - 1
[ ] D: y = 18x + 3

Solution: At x = 0 , y = 1^3 = 1 . Derivative: frac{dy}{dx} = 36x+1^2 cdot 6 = 186x+1^2 . At x = 0 , slope m = 18 . Equation: y - 1 = 18x - 0 implies y = 18x + 1 . [M1][A1]

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**Q6.** Find the equation of the tangent line to the curve y = 7x + 1^3 at the point where x = 0 .

[ ] A: y = 21x - 1
[ ] B: y = 7x + 1
[ ] C: y = 21x + 3
**[x] D: y = 21x + 1**

Solution: At x = 0 , y = 1^3 = 1 . Derivative: frac{dy}{dx} = 37x+1^2 cdot 7 = 217x+1^2 . At x = 0 , slope m = 21 . Equation: y - 1 = 21x - 0 implies y = 21x + 1 . [M1][A1]

**Q7.** Find the equation of the tangent line to the curve y = 8x + 1^3 at the point where x = 0 .

[ ] A: y = 8x + 1
[ ] B: y = 24x + 3
[ ] C: y = 24x - 1
**[x] D: y = 24x + 1**

Solution: At x = 0 , y = 1^3 = 1 . Derivative: frac{dy}{dx} = 38x+1^2 cdot 8 = 248x+1^2 . At x = 0 , slope m = 24 . Equation: y - 1 = 24x - 0 implies y = 24x + 1 . [M1][A1]

**Q8.** Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 text{ cm/s} . What is the rate of change of volume frac{dV}{dt} when the radius is r = 3 text{ cm} ?

**[x] A: 18pi text{ cm}^3/text{s}**
[ ] B: 36pi text{ cm}^3/text{s}
[ ] C: 9pi text{ cm}^3/text{s}
[ ] D: frac{18}{2}pi text{ cm}^3/text{s}

Solution: V = frac{4}{3}pi r^3 implies frac{dV}{dt} = 4pi r^2 frac{dr}{dt} . At r = 3 , frac{dV}{dt} = 4pi 3^2 0.5 = 2pi 9 = 18pi text{ cm}^3/text{s} . [M1][A1]